

Measuring Solutions

There are many different ways to describe solutions quantitatively. All methods describe the amount of solute present compared to the amount of solvent present in a solution.

The usefulness of each method is situation-specific and dependent upon the state of the solute, the concentration of the solute and the intended audience/application. Some methods that are commonly used in industrial chemistry are impractical for a small-scale chemistry lab, while more theoretical approaches are useful in understanding the nature of the solution, but are not very helpful in preparation.

1. Solution Concentration

The most common measure of concentration in science labs expresses how much solute a solution contains by stating the amount of moles of solute present per litre of solution. This is referred to as **molarity**. It is expressed as mol/L or M.

Molarity is the most useful method in solution stoichiometry calculations, as the solute is moles and thus greatly simplifies the calculations.

Since 1 mol/L contains 1 mole of solute per litre of solution, we can formulate the following equation:

$$M = \frac{n}{V}$$

Where M is the concentration in mol/L. This unit can also be expressed with a capital "M". So, M can refer to both the variable of molarity and the unit of measurement! Note that the textbook uses different variables...be careful!

Example 1: What mass of NaCl is needed to make 250 mL of a 0.850 M solution?

Given:

$$M = 0.850 \text{ mol/L}$$

$$V = 250 \text{ mL} = 0.250 \text{ L} \quad (\text{convert so that units are consistent})$$

$$n = MV = (0.850 \text{ mol/L})(0.250 \text{ L}) = 0.212 \text{ mol}$$

$$m = nMM \quad (\text{don't confuse molar mass (MM) with molarity (M)})$$

$$m = (0.212 \text{ mol})(58.5 \text{ g/mol}) = 12.4 \text{ g}$$

Therefore, 12.4 g of NaCl is needed to make 250 mL of a 0.850 M NaCl solution

Practice Questions:

1. How many moles of KOH is needed to prepare 250 mL of a 0.5 M solution? **(0.125mol)**
2. 0.7 g of NaOH is placed in 2.0 L volumetric flask of water. What is the final concentration of the solution? **(0.009 M)**
3. What is the concentration in moles of a solution that contains 100 g of Na_2SO_4 in 10.0 L of solution? **(0.07M)**
4. What volume of 0.14 M HCl would contain 5.0 g of HCl? **(978mL)**
5. How would you prepare 500 mL of a 0.02 M NaOH solution? **(0.4g in 500mL H₂O)**

Textbook Questions:

p. 284 #8

p. 287 # 14-17

p. 290 # 19-22

p. 290 3, 7-10

2. Dilutions

Commercially available concentrated solutions are called stock solutions. All solutions are shipped and stored as stock solutions which means they are as concentrated and as pure as can be to save space and money. For example, stock HCl is 12.0 mol/L. Typically, stock solutions are diluted for lab use.

Consider: 10.0 mL of a 12.0 M HCl

$$M_1 = 12.0 \text{ mol/L}$$

$$V_1 = 10\text{mL} = 0.010\text{L}$$

$$n = M_1 V_1 = (12.0 \text{ mol/L})(0.010\text{L}) = 0.12 \text{ mol}$$

- If we add water to dilute the stock solution, the solution volume will change and; therefore, so will the concentration, M ($M = n/V$). We will identify the new volume and concentration by different subscripts: (M_2 , V_2)
- However, the moles, n , will not change as we did not add or remove any moles of HCl.
- Therefore, after dilution, $n = M_2 V_2$
- Since n is constant, $M_1 V_1 = n = M_2 V_2$

This result is referred to as the **dilution equation**:

$$M_1 V_1 = M_2 V_2$$

Example 2.

What volume of 16.0 M nitric acid solution is needed to prepare 500 mL of a 0.500 M nitric acid solution?

Given:

$$M_1 = 16.0 \text{ mol/L}$$

$$M_2 = 0.500 \text{ mol/L}$$

$$V_1 = ?$$

$$V_2 = 500 \text{ mL} = 0.500 \text{ L}$$

$$M_1 V_1 = M_2 V_2$$

$$V_1 = \frac{M_2 V_2}{M_1}$$

$$V_1 = \frac{(0.500\text{mol/L})(0.500\text{L})}{(16.0\text{mol/L})}$$

$$V_1 = 0.015625\text{L} = 15.6\text{mL}$$

Therefore, to make desired solution, 15.6 mL of 16.0 M nitric acid solution is used

1. What volume of 12 molar acid would be needed to make 500 mL of a 4.0 molar solution? **(166.67mL)**
2. To 5 mL of a 1.0 M solution 45 mL of water was added. What is the concentration of the new solution? **(0.1M)**
3. What volume of 18.0 M H_2SO_4 solution would contains 0.5 moles of H_2SO_4 ? **(27.8mL)**
4. A concentrated ammonia (NH_3) solution is 28.5 % ammonia by mass and has a relative density of 0.90 g/mL. Calculate the concentration of the solution. **(15.1M)**

Textbook Questions:

p.302 #1,2,4a,5a
p.309 #9,11,17a
p.306 #6-8, 3-5
p.310 #12,13,18a

Reactions in Solutions Questions

Part A: Answers at end of doc

1. Show the dissociation of the following substances as a balanced chemical equation:

- | | |
|--|---|
| a) KCl (s) | f) CCl ₄ (l) |
| b) MgF ₂ (s) | g) Ag ₂ S (s) |
| c) FeS (s) | h) NH ₄ Cl (s) |
| d) Hg ₂ SO ₄ (s) | i) (NH ₄) ₂ Cr ₂ O ₇ (s) |
| e) Ca ₃ (PO ₄) ₂ (s) | j) K ₂ CrO ₄ (s) |

2. Write the net ionic equation for the dissociation of: **Answers at end of doc**

- a) ammonium sulfate in water
b) iron (III) carbonate in water

3. Calculate the number of moles of all aqueous ions in the following solutions, assuming each dissolved substance dissociates completely in solution.

- a) 0.60 L of 0.20 mol/L K₂SO₄ **n of K₂SO₄ = (0.20 mol/L) (0.60L) = 0.12 mol of compound, so 0,12 mol of SO₄ as there is only one but 2x 0.12 = 0.24 mol of K⁺ ions as there are two in the compound. Follow similar logic for below.**
- b) 0.450 L of 0.300 M Na₃PO₄ (Na ions = 0.405 mol , PO₄ ions = 0.135mol)
- c) 75.0 mL of 0.160 mol/L MnCl₂ (Mn ions = 0.012 mol, Cl ions = 0.024 mol)
- d) 0.950 L of 0.235 M Al₂(SO₄)₃ (Al ions = 0.4465 mol, SO₄ ions = 0.66975 mol)
- e) 25 mL of 0.065 M SnCl₄ (Sn ions = 0.01625 mol, Cl ions = 0.065 mol)

4. A student has 250 mL of a 0.35 M CuCl₂ solution. What volume of 0.15 M solution can be made from this? How much water must be added?

5. Water is added to 100 mL of 6.0 M KOH solution until the final volume is 1.5 L. What is the molarity of the new diluted solution?

Part B: Solution Stoichiometry

- 400.0mL of a 0.800 mol/L HCl solution is poured over excess zinc metal. Calculate the mass of hydrogen produced. **(0.320g)**
- 35mL of a 0.25M NaOH solution is required to neutralize 20mL of a HCl solution. Calculate the concentration of the acid. **(0.44M)**
- A 100mL of 1.50M HCl is required to neutralize a sample of 1.40M Ca(OH)₂ solution. Calculate the volume of the base. **(53.6mL)**
- The following reaction is conducted: FeCl_{3(aq)} + NaOH_(aq) → Fe(OH)_{3(s)} + NaCl_(aq) If excess NaOH was reacted with 260.0 mL of FeCl₃ to produce 3.50 g of Fe(OH)₃, what was the initial concentration of the FeCl₃ solution? **(0.126M)**
- If 3.05 g of sodium is added to 100.0 mL of distilled water, what concentration of NaOH will be produced (hydrogen gas is also formed)? **(1.33M)**

- In the following reaction: $K_{(s)} + HCl_{(aq)} \rightarrow KCl_{(aq)} + H_{2(g)}$ a 250.0 mL container is full of HCl while 3.0 g of K is added. What is the concentration of KCl formed? **(0.31M)**
- The following reaction was conducted: $BaCl_{2(aq)} + Na_2SO_{4(aq)} \rightarrow BaSO_{4(s)} + NaCl_{(aq)}$ Excess barium chloride was reacted with 60 g of a 10% by mass sodium sulphate solution. What mass of barium sulphate will be collected? **(9.86g)**
- Ethanol (C_2H_5OH) is converted into ethylene ($H_2C=CH_2$) and water. The typical percentage yield of ethylene is 82%. If 50 g of ethylene was obtained, what concentration of ethanol solution was used? **(2.2 mol/L)**
- A 150 mL solution of 0.6 mol/L unknown acid, H_nA , was used to neutralize 900 mL of 0.15 M $Ba(OH)_2$ solution. Determine the value for n in the acid. **(n = 3)**

3. Solution Percentages

One common industrial method relies on mass fraction of a solution which is a measure of the amount of solute compared to amount of solvent present in a solution. These are typically used to quickly prepare solutions. There are three traditional methods used to do this.

1. % V/V

$$= \frac{\text{volume of solute}}{\text{volume of solution}} \times 100\%$$

where the volume of the solution = (volume of solute + volume of solvent)

What is the % V/V of a solution prepared by adding 10 mL of solute to 45 mL of solvent
 $= 10/(10 + 45) \times 100\% = 18.2\% \text{ V/V}$ (include this term in your answer)

2. % W/V

$$= \frac{\text{mass of solute}}{\text{volume of solution}} \times 100\%$$

where the volume of the solution = (volume of solute + volume of solvent). However the solute mass does not change the volume of the solvent to any real extent so we ignore this volume so the equation becomes

$$= \frac{\text{mass of solute}}{\text{volume of solvent}} \times 100\%$$

What is the % W/V of a solution prepared by adding 10 g of solute to 45 mL of solvent
 $= 10/45 \times 100\% = 22.2\% \text{ W/V}$ (include this term in your answer)

3. % W/W

$$= \frac{\text{mass of solute}}{\text{mass of solution}} \times 100\%$$

where the mass of the solution = (mass of solute + mass of solvent)

note that we assume the density of water = 1.0 g/mL

What is the % W/W of a solution prepared by adding 10 g of solute to 45 mL of water
 $= 10/(10 + 45) \times 100\% = 18.2\% \text{ W/W}$ (include this term in your answer)

Textbook Questions:

p. 284 #2-4

p. 287 #11,12

p. 309 #5-7

4. Extremely Small Concentrations

For certain solutions in aqueous chemistry and biochemistry, solutions often describe the amount of a particular molecule, element or toxin. In many of these applications, the % of solute is so incredibly small that it is impractical to communicate using percentages.

Instead, we look at the mass of solute per mass of solution as a ratio. Statistics are most often described in this manner; for example mortality rates are described in deaths per 100,000 people. (Canada's mortality rate is 7.74; USA's is 8.38).

- In chemistry, we need considerably smaller measurements such as **parts per million (ppm)** or **parts per billion (ppb)** or **parts per trillion (ppt)**
- 1 ppm equals 1 g of solute in 1 million g of solution (the solution is 1 million times bigger).
- Since there is so little solute compared to amount of solution, we really just consider the amount of solute/amount of solvent; therefore, 1 ppm = 1 g of solute/1 million mL of water (remember 1 mL = 1 g of water).
- As long as the difference between the amount of solute and the amount of solvent is 1 million, it is a ppm; therefore, 1 ppm = 1 mg/kg (1 kg = 1 million mg) or 1 µg/g (1 g = 1 million µg).
- Likewise, 1 ppb = 1g/1 billion mL of water (the solution is 1 billion times bigger).

Although this method is useful for describing extremely small concentrations, there is a little extra work in doing quantitative calculations, as we must first ensure that the **units** of the solute and solvent are the same before working out the stoichiometry.

Example 1.

In Ontario, drinking water should have less than 10 ppb of lead in it. What is the maximum amount of lead allowed in a 500 mL bottle of water?

1mL of water = 1g of water

10ppb = 10g in 1,000,000,000g of water

Set up equivalent ratios:

If 10g of lead	Then how many g
In 1×10^9 g of water	In 500g of water?

$$\frac{10g}{1 \times 10^9 g} = \frac{x}{500g}$$

$$x = 5 \times 10^{-6} g = 5 \mu g$$

Example 2.

A 30 g sample of herbal medicine was recently banned because it was found to contain 0.0705 mg of strychnine (a powerful naturally occurring poison). What was this toxic level in ppm?

1 ppm = 1 g in 1,000,000g sample = 1×10^6 g

0.0705mg = 7.05×10^{-5} g

If 7.05×10^{-5} g toxin	Then how many g
in a 30g sample	in a 1×10^6 g sample?

$$\frac{7.05 \times 10^{-5} g}{30 g} = \frac{x}{1 \times 10^6 g}$$

$$x = 2.35 g$$

Therefore, there is 2.35g of toxin per million grams of medicine, or 2.35ppm

Picturing Extremely Small Concentrations:

Measurement	Equivalent Ratio
1 ppm	1 drop of water in a full bathtub 1 second in 11.5 days 1 minute in 2 years
1 ppb	1 drop of water in a full swimming pool 1 square of toilet paper in a line stretching from New York to London, England 1 second in 32 years 1 pinch of salt in 10160.5kg of potato chips

Textbook Questions:

p. 284 #5-7

p. 309 #10

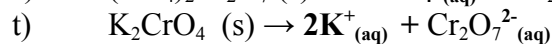
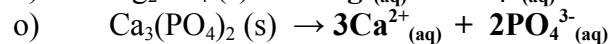
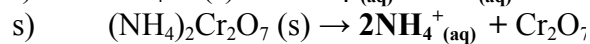
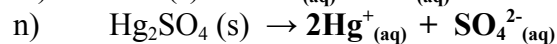
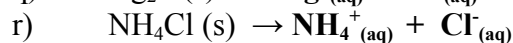
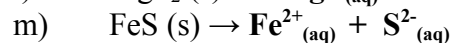
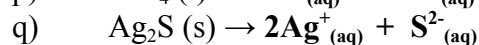
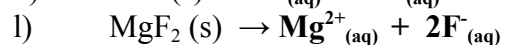
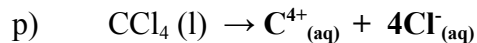
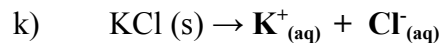
p. 287 #13

p. 290 #2,4-6

ANSWERS: Reactions in Solutions Questions

Part A:

6. Show the dissociation of the following substances as a balanced chemical equation:



7. Write the net ionic equation for the dissociation of: - With only ONE solid dissolving, the net equation is the same as the dissociation equations above

